AccessLens v2: Policy Access Risk Simulation, Eligibility Discovery and Explainable AI Interface

Project Progress & Update Report

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Abstract

This report documents the progress and evolution of AccessLens v2, a policy research system designed to model non-financial barriers in accessing government welfare schemes. The system has evolved from a simulation-only tool to a fully integrated decision-support pipeline. Key achievements include the creation of enhanced personas with realistic attributes, a new eligibility-aware scheme schema, the Persona Mapping Function for privacy-preserving user input handling, the Eligibility Discovery Module for rule-based scheme matching, and full pipeline integration combining eligibility assessment with access risk simulation. The current system accepts user inputs, maps them to synthetic personas, identifies potentially eligible schemes, computes access difficulty scores, and generates structured recommendations with the novel Access Gap metric. This report outlines current capabilities, key innovations, limitations, and the next development phase.

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1 Introduction

AccessLens v2 is a policy research system designed to model and simulate non-financial barriers that prevent eligible individuals from accessing government welfare schemes. The system addresses a critical gap in policy analysis: while eligibility criteria determine who qualifies for benefits, non-financial barriers such as lack of awareness, documentation complexity, digital inaccessibility, and institutional dependency often determine who actually receives them.

Unlike predictive AI systems that operate as black boxes, AccessLens v2 employs a deterministic, rule-based approach combined with counterfactual analysis. The system prioritizes transparency, reproducibility, and privacy preservation. AI components are used only for post-hoc explanation, not for decision-making or eligibility determination.

This report documents the evolution of AccessLens v2 from a conceptual simulation tool to a fully integrated decision-support pipeline. It describes what has been built, current system capabilities, key innovations, and the planned next development phase.

2 System Evolution

AccessLens v2 has progressed through three distinct phases, each building on the foundation of the previous phase.

2.1 Phase 1: Foundation Datasets

The initial phase focused on creating structured datasets to support barrier analysis:

- `schemes_v0`: Basic scheme information including domain, application mode, document requirements, and access characteristics.
- `personas_v0`: Synthetic population profiles with literacy level, digital access, document completeness, and institutional dependency attributes.
- `barriers_v0`: Taxonomy of four barrier types (awareness, documentation, digital, institutional) with associated severity values.

2.2 Phase 2: Simulation Core

The second phase implemented the core simulation capabilities:

- Deterministic access risk model for computing barrier-based friction scores.
- Counterfactual simulation engine for policy intervention analysis (awareness improvement, documentation simplification, assisted access removal, digital enablement).
- Batch simulation pipeline for systematic evaluation across scheme-persona combinations.
- Visualization outputs including risk heatmaps and intervention ranking.

2.3 Phase 3: Decision-Support Integration (Current Major Update)

The current phase represents a significant evolution from a simulation tool to a comprehensive decision-support system:

- Creation of `personas_v1` with realistic demographic and eligibility-related attributes.
- Definition and implementation of `schemes_v1` schema incorporating both eligibility criteria and access-related attributes.

- Development of the Persona Mapping Function (PMF) for privacy-preserving user input handling.
- Implementation of the Eligibility Discovery Module (EDM) for rule-based scheme matching.
- Full pipeline integration combining eligibility discovery with access risk simulation.
- Introduction of the Access Gap metric for actionable policy insights.

3 Dataset Enhancements

Significant enhancements have been made to the datasets to support eligibility discovery alongside access risk simulation.

3.1 Personas v1: Realistic Attributes

The persona dataset has been substantially upgraded from barrier-focused attributes to include realistic demographic and eligibility-relevant characteristics:

Table 1: Personas v1 Attribute Categories

Category	**Attributes**	**Description**
Demographic	age, gender, income_category	Basic demographic and economic status
Geographic	rural_urban	Urban or rural location
Occupational	occupation	Farmer, laborer, self-employed, salaried, student, unemployed, retired
Eligibility Flags	student_status, farmer_status, disability_status	Special category indicators
Barrier Attributes	digital_access, document_completeness, institutional_dependency	Retained from v0 for risk modeling

These enhancements enable the system to determine which schemes a persona is eligible for, while retaining barrier simulation capabilities.

3.2 Schemes v1: Eligibility + Access Schema

A new schema has been defined for schemes, incorporating both eligibility criteria and access-related attributes:

The dataset is currently partially populated for development purposes, with some fields marked as "not specified." This allows testing of the pipeline while acknowledging data quality limitations.

3.3 Barriers Dataset

The barrier taxonomy remains stable with four barrier types (awareness, documentation, digital, institutional) and associated severity values. No major changes were required for this phase.

4 Persona Mapping Function (PMF)

The Persona Mapping Function is a critical component that enables the system to accept arbitrary user inputs while preserving privacy and maintaining deterministic behavior.

Table 2: Schemes v1 Schema Categories

Category	Attributes	Description
Eligibility Criteria	age_min, age_max, gender, income_category, occupation, rural_urban, student_eligible, farmer_eligible, disability_eligible	Conditions for scheme qualification
Access Attributes	application_mode, documents_required_count, digital_only, assisted_access_available, steps_count	Barriers and friction points

4.1 Purpose

The PMF maps real user-provided attributes to the closest synthetic persona from the `personas_v1` dataset. This design ensures:

- No storage of real user data (privacy preservation)
- Reproducible and deterministic mapping
- Ability to handle partial or missing inputs gracefully
- Consistent downstream processing using synthetic personas

4.2 Implementation

The mapping uses a weighted similarity scoring system:

$$\text{Similarity}(U, P) = \frac{\sum_{i=1}^{n} w_i \times \text{match}_i(U_i, P_i)}{\sum_{i=1}^{n} w_i \times \text{provided}_i(U_i)}$$

Where:

- U represents user input attributes
- P represents persona attributes
- w_i are attribute-specific weights
- match_i returns 1 for exact match, 0 otherwise (with special handling for numeric age ranges)
- provided_i is 1 if the user provided attribute i, 0 otherwise

The function handles missing inputs by normalizing the similarity score based only on attributes that were provided. This allows flexible user interaction while maintaining deterministic behavior.

4.3 Output

The PMF returns:

- `persona_id`: The matched synthetic persona identifier
- `matched_attributes`: List of attributes used in matching
- `confidence`: Normalized similarity score (0-1)

5 Eligibility Discovery Module (EDM)

The Eligibility Discovery Module identifies which welfare schemes a given persona is likely eligible for. It is a rule-based, deterministic system that operates without machine learning.

5.1 Purpose

The EDM enables users to discover potentially applicable schemes based on their characteristics. It explicitly avoids making official eligibility determinations and clearly communicates this limitation.

5.2 Implementation

The module matches scheme eligibility criteria against persona attributes using a weighted scoring system:

$$\text{EligibilityScore}(S, P) = \frac{\sum_{j=1}^{m} w_j \times \text{meets}_j(S_j, P_j)}{\sum_{j=1}^{m} w_j \times \text{active}_j(S_j)}$$

Where:

- S represents scheme eligibility criteria
- P represents persona attributes
- meets_j returns 1 if the criterion is met, 0 otherwise
- active_j is 1 if the scheme specifies the criterion, 0 otherwise
- w_j are attribute-specific weights

5.3 Key Features

The EDM incorporates several innovative features:

- **Handles "not specified" values**: Both scheme criteria and persona attributes can be unspecified; the module adapts accordingly without breaking.
- **Dynamic weight normalization**: Weights are normalized based on active criteria only, ensuring fair scoring across schemes with varying levels of detail.
- **Minimum active fields constraint**: Schemes with insufficient specified criteria receive lower scores, reflecting uncertainty.
- **Data quality penalty**: Schemes with many "not specified" fields are penalized to reflect real-world data limitations.

5.4 Output

The EDM returns a ranked list of schemes with:

- `scheme_id` and `scheme_name`
- `eligibility_score` (0-1, normalized)
- `eligibility_label`: High (score ≥ 0.7), Medium (0.4–0.7), Low (< 0.4)

6 Access Risk Model Integration

The access risk model from Phase 2 has been integrated into the pipeline. It computes non-financial access difficulty for a given scheme-persona pair.

6.1 Model Overview

The deterministic formula remains:

$$\text{RiskScore} = \frac{\sum_{i=1}^{n} \left(\text{Severity}_i \times \text{Activation}_{i,\text{scheme}} \times \text{Amplification}_{i,\text{persona}}\right)}{\sum_{i=1}^{n} \text{Severity}_i}$$

Where:

- Severity_i is the baseline severity of barrier i (0-1 scale)
- $\text{Activation}_{i,\text{scheme}}$ is 1 if barrier i applies to the scheme, 0 otherwise
- $\text{Amplification}_{i,\text{persona}}$ captures how persona attributes affect susceptibility to barrier i (0-1 scale)

6.2 Current Integration Status

- The model successfully computes risk scores for scheme-persona pairs.
- For schemes with incomplete data in `schemes_v1`, some barrier activation values are defaulted based on available attributes.
- Risk scores are normalized to the 0-1 range, with higher values indicating greater access difficulty.

6.3 Limitation Acknowledgment

The current implementation recognizes data gaps. When scheme attributes are "not specified," the risk model either uses reasonable defaults or marks the result as having reduced confidence. Full alignment with `schemes_v1` schema is planned for the next phase.

7 Full Pipeline Integration

The most significant achievement in this phase is the full integration of all components into a complete, working pipeline that transforms user inputs into actionable recommendations.

7.1 Pipeline Flow

The system now operates as follows:

1. **User Input**: User provides attributes such as age, income category, occupation, location, and optional barrier-related information.
2. **Persona Mapping (PMF)**: The input is mapped to the closest synthetic persona from `personas_v1`.
3. **Eligibility Ranking (EDM)**: The matched persona is evaluated against all schemes to determine eligibility scores.
4. **Top-K Selection**: The top k schemes (default $k = 10$) are selected based on eligibility score.

5. **Access Risk Calculation**: For each selected scheme, access risk scores are computed.
6. **Final Output**: A structured recommendation is generated, combining eligibility and risk information.

7.2 Access Gap

A key innovation in this integration is the **Access Gap** metric:

$$\text{AccessGap} = \text{EligibilityScore} - \text{RiskScore}$$

Interpretation:

- **High eligibility + High risk** (small or negative gap): The person qualifies for the scheme but faces significant non-financial barriers. This indicates an *access problem* requiring intervention.
- **High eligibility + Low risk** (large positive gap): The person qualifies and faces few barriers. This is the *ideal case*.
- **Low eligibility**: The person likely does not qualify; access risk is less relevant, though still informative.

The Access Gap provides a simple, intuitive way to prioritize policy interventions: schemes with high eligibility but high risk are candidates for barrier-reduction efforts.

8 Current System Capabilities

AccessLens v2, in its current integrated form, provides the following capabilities:

- **Flexible user input**: Accepts partial information (age, income category, occupation, location, and optional attributes) without requiring complete data.
- **Privacy-preserving mapping**: Maps user inputs to synthetic personas without storing any personal data.
- **Eligibility discovery**: Identifies potentially applicable schemes based on rule-based matching of eligibility criteria.
- **Risk assessment**: Computes access difficulty scores considering four barrier types (awareness, documentation, digital, institutional).
- **Integrated recommendations**: Combines eligibility and risk into structured output with the Access Gap metric.
- **Deterministic and explainable**: All calculations are rule-based, reproducible, and can be traced back to inputs.
- **Modular design**: Components (PMF, EDM, Risk Model) can be tested, validated, and improved independently.

The fully working pipeline has been implemented and tested with sample inputs, demonstrating the feasibility of the integrated approach.

9 Key Innovations

Several innovations distinguish AccessLens v2 from existing approaches to welfare access analysis:

- **Separation of eligibility and access risk**: Most systems consider eligibility only; AccessLens v2 explicitly models the gap between qualifying for a scheme and actually accessing it.
- **Access Gap metric**: Provides a simple, actionable indicator for prioritizing policy interventions. Schemes with high eligibility but high risk are clear targets for barrier-reduction efforts.
- **Data quality-aware scoring**: The EDM penalizes schemes with incomplete eligibility criteria, reflecting real-world data limitations and providing more realistic outputs.
- **Deterministic + explainable architecture**: No black-box machine learning models in core decision-making; all outputs can be traced to inputs and rules.
- **Privacy-preserving persona mapping**: User data is never stored; mapping uses synthetic personas with no retention, addressing privacy concerns common in welfare contexts.
- **Handling of "not specified" values**: Both PMF and EDM gracefully handle missing or unspecified values, making the system robust to real-world data limitations.

10 Limitations (Updated)

The current system has several limitations that are being addressed in the next development phase:

- **Incomplete scheme data**: `schemes_v1` is partially populated; many eligibility and access fields are marked "not specified," which reduces precision and confidence in outputs.
- **Risk model alignment**: The access risk model was originally designed for `schemes_v0` and requires full alignment with the enhanced `schemes_v1` schema to leverage all available data.
- **Eligibility suggestions are not official**: The system explicitly states that outputs are for exploration and research purposes only. All eligibility determinations require verification through official government channels.
- **No real-world validation**: Results have not been validated against actual access patterns, official eligibility determinations, or field observations.
- **Limited scheme coverage**: The current dataset includes a modest number of schemes; broader coverage across domains (health, education, agriculture, social protection) is needed for comprehensive analysis.
- **No AI explanation layer yet**: AI-driven natural language explanations, translation, and query interfaces are planned but not yet implemented.
- **No interactive dashboard**: The system currently operates as a command-line or programmatic interface; a user-friendly web interface is planned.

11 Next Development Phase

The following priorities have been identified for the next development phase:

- **Complete schemes_v1 dataset**: Expand from current coverage to 100+ schemes across multiple domains (health, education, agriculture, social protection, housing, livelihoods). Ensure all eligibility and access fields are populated with high-quality data from official sources.
- **Align risk model with schemes_v1**: Update the access risk model to fully utilize the enhanced scheme schema, ensuring accurate barrier activation based on scheme attributes.
- **Implement AI explanation layer**: Integrate large language models to convert simulation outputs into natural language explanations that are accessible to non-technical stakeholders.
- **Add multilingual translation**: Support for Hindi, Marathi, Tamil, Bengali, and other Indian languages to improve accessibility for diverse users.
- **Develop natural language query interface**: Allow users to ask questions like "Which schemes am I eligible for?" and "What barriers would I face?" and receive structured responses.
- **Build Streamlit dashboard**: Create an interactive web interface with:
  - Form-based user input for demographic and eligibility information
  - Visual display of eligible schemes with eligibility scores
  - Access risk visualization with barrier breakdowns
  - Counterfactual simulation controls for policy exploration
  - Export functionality for results
- **Conduct user testing**: Evaluate the system with policy practitioners, civil society organizations, and potential beneficiaries to refine design, interface, and explanation quality.
- **Documentation and release**: Prepare comprehensive documentation and consider open-source release to enable broader use and contribution.